Notes: Zwiebel (AER, 1996)

Dynamic Capital Structure under Managerial Entrenchment

Contents

1 Big picture

- **Question.** Can debt still discipline managerial empire-building when managers themselves choose capital structure and can therefore undo leverage decisions made earlier? Zwiebel's answer is yes, but only if debt works through a *dynamic control mechanism*: bankruptcy weakens managerial entrenchment and raises the chance of replacement.
- **Core idea.** Managers enjoy both control and expansion. They would like to overinvest in bad projects, but they also want to avoid takeovers. Debt is useful because it makes bad investment choices more likely to trigger bankruptcy, and bankruptcy reduces entrenchment. Managers therefore voluntarily choose debt as a *self-imposed commitment device*.
- **Why this matters.** Standard free-cash-flow stories usually take debt as imposed *ex ante* by some discipliner (investors, creditors, raiders, or the market), but that creates a dynamic-consistency problem: why would managers not later reverse the capital structure? Zwiebel's model solves that problem by making leverage the optimal choice of partially entrenched managers themselves.
- **Mechanism in one sentence.** Debt is not useful because it mechanically deprives managers of start-up cash for bad projects; it is useful because it increases the probability that bad projects lead to bankruptcy and loss of control.
- **Main theoretical result.** In each period, managers choose the *minimum net debt* necessary to deter a takeover by credibly promising enough future efficiency. They do this by committing to reject bad projects in exactly those periods in which unconstrained empire-building would otherwise make them vulnerable.
- **Main dynamic result.** The maturity structure of debt emerges endogenously. Some types issue no debt, some issue only short-term debt, some only longer-term debt, and some both. Better investment opportunities imply less debt and less frequent debt payments.
- **Main dividend result.** Debt and dividends are jointly chosen. Because what matters is *net debt*, managers who want debt to bite must also disgorge cash. This gives a dynamically consistent reason why empire-building managers may voluntarily pay dividends.
- **Main "cost of debt" result.** In this model, excessive debt is not costly because it blocks good projects. Good projects are always undertaken. Excessive debt is costly because once bankruptcy becomes inevitable, debt no longer disciplines bad projects at all.
- **Key empirical implications.**
  - leverage is lower when good investment opportunities are better;
  - debt frequency and debt size fall with growth opportunities and rise in mature or declining industries;
  - leverage and dividends should be positively coordinated;
  - bankruptcy and leverage rise as a manager approaches the end of tenure, except in endgame periods where discipline breaks down;
  - firms near distress may exhibit *more* empire-building, not less.
- **Economic takeaway.** The paper is a dynamic agency theory of capital structure. Firms use debt not only to move cash out of managers' hands, but to change what managers fear about the future.

2 Setup and economic objects (key objects)

2.1 Firm, cash flows, and projects

- The baseline model has **three periods**. Assets in place generate deterministic cash flow y each period, and have no value after period 3.
- In each period the manager may undertake **one additional project**. With probability t, a good project is available; otherwise only a bad project is available. The parameter t is public information and indexes the firm's investment opportunities or, equivalently, the manager's type.
- Good and bad projects have deterministic net returns in the baseline:

$$r_G = r < y, \qquad r_B = -r. \tag{A}$$

- Projects require **no initial outlay**. This is deliberate: the paper wants to rule out the usual "debt blocks good projects because the firm lacks cash" mechanism. Good projects are always feasible.

Interpretation. Because bad projects do not require cash up front either, debt can only matter through how it changes future bankruptcy risk and therefore future control.

2.2 Managerial preferences

- The manager receives utility $A > 0$ from **retaining control** of the firm.
- The manager receives utility $B > 0$ from **undertaking an additional project**, regardless of whether that project is good or bad.
- Utility from control and project-taking is additive and time-independent.

Interpretation. A is the value of running the empire; B is the taste for expansion. The model does not assume anything special about the relative sizes of A and B.

2.3 Entrenchment, takeover, and bankruptcy

- Outside bankruptcy, a manager can only be removed if a takeover increases firm value by more than an entrenchment amount e.
- The paper scales entrenchment by the size of the project's return wedge:

$$E \equiv \frac{e}{r}. \tag{B}$$

- In the baseline analysis, Zwiebel assumes $E < 2$ so that entrenchment is not too large relative to the cost of two bad projects.
- Bankruptcy **removes entrenchment**. If the firm defaults, it is reorganized efficiently, and the incumbent manager is replaced whenever replacement raises value.
- The replacement manager is assumed to have **no access to new projects**. This simplifies the continuation values and makes the outside option a disciplined manager who simply runs assets in place.

Interpretation. The model's entire force comes from the asymmetry between normal times and bankruptcy. Outside bankruptcy, entrenchment protects the incumbent. In bankruptcy, that protection disappears.

2.4 Capital structure, dividends, and net debt

Let D_i denote debt due in period i, $w_i(D_i)$ its market value, d_i dividends at date i, and L_i cash on hand before capital-structure decisions. Cash after those decisions is

$$\bar{L}_i = L_i + w_i(D_i) - d_i. \tag{C}$$

The key state variable is **net debt**:

$$\bar{D}_i \equiv D_i - \bar{L}_i. \tag{D}$$

Interpretation.

- $\bar{D}_i$ is the amount that must be covered by operating and project cash flow to avoid bankruptcy in period i.
- This is why dividends matter: if the manager keeps cash inside the firm, debt may no longer bite.
- All substantive leverage results are therefore about *net debt*, not gross debt.

2.5 Timeline (Figure 1)

Each period follows this order:

1. manager chooses debt issuance/repurchase and dividends;
2. market for corporate control operates;
3. availability of a good project is learned;
4. manager chooses whether to invest;
5. returns are realized and debt is serviced;
6. if debt cannot be paid, bankruptcy and efficient reorganization occur.

Interpretation. The order matters. Capital structure must be chosen before the takeover decision if debt is to work as a credible commitment device.

3 Model and equilibrium characterization (all equations + interpretation)

3.1 Final-period takeover condition

In period 3 there is no future left to protect, so the manager undertakes any available project. Equity value with the incumbent in control is

$$v_3(t, L_3, 0) = L_3 + y + (2t - 1)r, \tag{1}$$

while with a replacement manager it is

$$\bar{v}_3(L_3, 0) = L_3 + y. \tag{2}$$

Therefore the manager's value contribution is

$$V_3(t) = v_3(t) - \bar{v}_3 = (2t - 1)r. \tag{3}$$

A takeover occurs if $V_3(t) < -e$, that is,

$$t < \underline{t} \equiv \frac{1 - E}{2}. \tag{4}$$

Interpretation. In the final period, only managers with enough chance of a good project are worth keeping. Since debt cannot affect future behavior anymore, it serves no purpose in period 3.

3.2 Period 2: debt as one-period commitment

If no debt is due in period 2, the manager invests in all projects, so continuation value at the beginning of period 2 is

$$V_2(t) = 2(2t - 1)r. \tag{5}$$

Hence a second-period takeover occurs whenever

$$2(2t - 1)r < -e \quad \Longleftrightarrow \quad t < \frac{1}{2} - \frac{E}{4}. \tag{6}$$

For managers in the interval

$$\underline{t} \le t < \frac{1}{2} - \frac{E}{4}, \tag{7}$$

this takeover can be avoided by issuing net debt satisfying

$$y - r < \bar{D}_2 \le y. \tag{8}$$

This makes the firm default if and only if the manager undertakes a bad project in period 2. Because bankruptcy then causes replacement, the manager prefers to reject bad projects but still accepts good ones.

The resulting period-2 value contribution is

$$V_2(t) = \begin{cases} (4t - 2)r, & t \ge \dfrac{1}{2} - \dfrac{E}{4}, \\ (3t - 1)r, & \underline{t} \le t < \dfrac{1}{2} - \dfrac{E}{4}. \end{cases} \tag{9}$$

Interpretation.

- Without debt, these managers would enjoy two periods of unconstrained empire-building and would be taken over.
- With debt, they buy exactly one period of discipline: they give up bad projects in period 2, survive the control market, and then are unconstrained in period 3.
- The paper's logic is already visible here: the manager chooses the *minimum discipline* required to stay in office.

3.3 Full three-period equilibrium

Suppose first that the manager reaches period 1 with no binding debt. Using equation (9), the value contribution at the beginning of period 1 is

$$V_1(t) = \begin{cases} (6t-3)r, & t \geq \frac{1}{2} - \frac{E}{4}, \\ (5t-2)r, & \underline{t} \leq t < \frac{1}{2} - \frac{E}{4}. \end{cases} \tag{10}$$

Hence a period-1 takeover occurs if either

$$t < \frac{2}{5} - \frac{E}{5} \qquad \text{or} \qquad \frac{1}{2} - \frac{E}{4} \leq t < \frac{1}{2} - \frac{E}{6}. \tag{11}$$

These threatened types can avoid takeover by issuing first-period net debt

$$y - r < \bar{D}_1 \leq y, \tag{12}$$

which commits them to reject bad projects in period 1.

This yields the paper's main partition of managerial types.

3.4 Proposition 2: five regions of equilibrium behavior

Let the lower cutoff be $\underline{t} = (1-E)/2$. Then the unique subgame-perfect equilibrium in the three-period deterministic model is:

1. $t < \underline{t}$**.** The manager is taken over at the beginning of period 1 for any capital structure.
2. $\underline{t} \leq t < \frac{2}{5} - \frac{E}{5}$**.** The manager issues debt in both periods 1 and 2. He accepts only good projects in periods 1 and 2 and accepts all projects in period 3.
3. $\frac{2}{5} - \frac{E}{5} \leq t < \frac{1}{2} - \frac{E}{4}$**.** The manager issues debt only in period 2. Period 1 is unconstrained, period 2 accepts only good projects, and period 3 is unconstrained.
4. $\frac{1}{2} - \frac{E}{4} \leq t < \frac{1}{2} - \frac{E}{6}$**.** The manager issues debt only in period 1. Period 1 accepts only good projects, but periods 2 and 3 are unconstrained.
5. $t \geq \frac{1}{2} - \frac{E}{6}$**.** The manager issues no debt and accepts all projects in all periods.

On the equilibrium path there is **no bankruptcy**; equilibrium debt is priced at face value.

The beginning-of-period-1 value contribution is

$$V_1(t) = \begin{cases} (6t-3)r, & \frac{1}{2} - \frac{E}{6} \leq t, \\ (5t-2)r, & \frac{2}{5} - \frac{E}{5} \leq t < \frac{1}{2} - \frac{E}{6}, \\ (4t-1)r, & \underline{t} \leq t < \frac{2}{5} - \frac{E}{5}. \end{cases} \tag{13}$$

Interpretation.

- The lowest surviving types need debt in *both* early periods because they need the most commitment.
- Intermediate types need commitment only once.
- The striking region is $\frac{1}{2} - \frac{E}{4} \le t < \frac{1}{2} - \frac{E}{6}$: these relatively better managers do *not* need period-2 debt, but three fully unconstrained periods would make them vulnerable to an early takeover, so they choose period-1 debt only.
- Debt maturity is therefore not monotone in a naive way. It depends on which future periods require discipline.

3.5 From payment dates to maturity structure: the N-period extension

Let $\mathcal{D}(N,t)$ denote the number of periods in which a manager of type t has debt payments due in an N-period game. Proposition 3 shows

$$\mathcal{D}(N,t) = N - \operatorname{int}\left(\frac{E+Nt}{1-t}\right), \tag{14}$$

where $\operatorname{int}(\cdot)$ is the greatest integer less than or equal to its argument.

A manager has debt due in the kth-to-last period if and only if

$$0 \le (k-1) - \operatorname{int}\left(\frac{E+(k-1)t}{1-t}\right) \ne k - \operatorname{int}\left(\frac{E+kt}{1-t}\right). \tag{15}$$

For $t \le 1/2$,

$$\lim_{N\to\infty} \frac{\mathcal{D}(N,t)}{N} = \frac{1-2t}{1-t}. \tag{16}$$

Interpretation.

- Managers with worse opportunities (lower t) need more frequent commitment.
- Debt frequency is weakly decreasing in growth opportunities.
- In long-horizon problems, low-type managers repeatedly roll over into debt obligations at regular intervals; maturity structure is an endogenous response to agency problems, not an arbitrary choice layered on top of them.

3.6 Stochastic outcomes and equilibrium bankruptcy

Zwiebel then allows project outcomes to be noisy:

$$\tilde{r}_i = r_i + \tilde{\varepsilon}, \qquad r_G = r, \qquad r_B = -r, \tag{17}$$

with mean-zero noise $\tilde{\varepsilon}$ distributed according to cdf F.

The equilibrium structure of managerial types stays similar, but now bankruptcy can occur *on the equilibrium path* even when the manager only undertakes good projects.

Let $\hat{t}$ denote the smaller root of

$$t^2\big[-2F(\hat{D}_2 - y - r)\big] + t\big[5 + F(\hat{D}_2 - y - r)\big] + (E-2) = 0. \tag{18}$$

The key debt levels are pinned down implicitly by the bad-project bankruptcy probabilities:

$$F(D_1^B - y + r) = \frac{B}{A + tB + (A+B)[1 - tF(\hat{D}_2 - y - r)]}, \tag{19}$$

$$F(\hat{D}_2 - y + r) = \frac{B}{A+B}, \tag{20}$$

$$F(D_1^F - y + r) = \frac{B}{2(A+B)}. \tag{21}$$

Here:

- D_1^B is first-period debt for managers who also issue second-period debt,
- $\hat{D}_2$ is second-period debt,
- D_1^F is first-period debt for managers who issue debt only in period 1.

Interpretation.

- Equations (19)–(21) say managers choose debt so that a bad project raises bankruptcy risk just enough to make self-restraint optimal.
- The paper now gets precise debt levels rather than the debt intervals of the deterministic case.
- On-path bankruptcy probabilities are positive because even good projects can end badly once noise is introduced.

3.7 Comparative statics of debt in the stochastic model

The paper emphasizes three clean results.

- **Debt falls with control benefits** A**.** When managers value staying in office more, a smaller increase in bankruptcy risk suffices to deter bad projects.
- **Debt rises with empire-building benefits** B**.** When managers enjoy project-taking more, they need a stronger bankruptcy threat to behave.
- **Debt and bankruptcy rise over tenure for constrained managers.** Among managers who issue debt in both periods, second-period debt exceeds first-period debt because the manager has less future control to lose later. Hence discipline must become stronger over time.

The paper also shows that among first-period debt issuers, higher-type managers issue *less* debt and face lower bankruptcy risk because they have more future rents to protect.

4 What makes the argument credible (and where it is limited)

4.1 Why the model solves the dynamic-consistency problem

- In standard free-cash-flow stories, someone else imposes debt early, but that does not explain why managers cannot later lever down or refinance.

- Here managers choose debt themselves, precisely because it helps them remain in control.
- Since the debt choice is privately optimal for the incumbent at each date, there is no tension between ex ante optimality and ex post incentives.

4.2 What assumptions do the heavy lifting

- **Bankruptcy reduces entrenchment.** Without this, debt would not change the manager's continuation value enough to matter.
- **Capital structure is chosen before the raider acts.** The raider must be able to condition on leverage for debt to commit management.
- **Managers cannot freely add debt after investment but before repayment.** If they could, they might refinance away the bankruptcy threat. Zwiebel argues that debt covenants restricting future debt can restore commitment.
- **Good projects need no up-front cash.** This isolates the commitment channel from financing-constraint stories.

4.3 Main limitations

- Entrenchment e is taken as exogenous; the paper does not solve for optimal entrenchment or governance structure.
- The replacement manager is deliberately simple: he runs existing assets but brings no new projects.
- The model is theory-first. It produces empirical implications, but does not test them directly.
- Managerial preferences A and B are reduced-form. The paper does not microfound compensation, boards, or internal incentives.

4.4 How it differs from standard free-cash-flow theories

- In a free-cash-flow model, debt is costly because it may block *good* projects.
- In Zwiebel's model, debt is costly because if it is too high, bankruptcy becomes inevitable and managers stop fearing it; then debt no longer blocks *bad* projects.
- This means the benefit and cost of debt come from the *same agency channel*, which is why the model naturally generates an interior leverage choice.

4.5 Relation to the literature

The paper speaks directly to Jensen's free-cash-flow view and to models by Grossman and Hart, Stulz, Hart, Hart and Moore, Williamson, and Harris and Raviv. It is especially close in spirit to Grossman and Hart (1982), but adds the dynamic structure needed to study repeated leverage choice, payout policy, and maturity design. Zwiebel also draws a connection to Akerlof and Romer's "looting" idea: near distress, managers may engage in more inefficient activity because the future is no longer worth protecting.

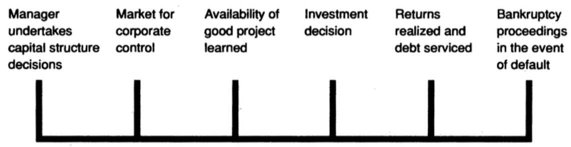

FIGURE 1. TIMELINE FOR EACH PERIOD

5 Figure 1 and key propositions

5.1 Figure 1: the timeline

Read. Figure 1 is the entire model in one line: capital structure first, control market second, project choice third, cash flow realization fourth, bankruptcy last.

Why it matters.

- Debt can commit only because the raider sees it before investment occurs.
- Bankruptcy can discipline only because repayment comes after the manager's project choice.
- The figure makes clear why the paper is about *dynamic* capital structure rather than a one-shot financing choice.

5.2 Proposition 1: the period-2 continuation equilibrium

Read. Proposition 1 solves the back end of the model. It identifies the threshold $\frac{1}{2} - \frac{E}{4}$ below which managers need second-period debt to survive.

Why it matters.

- It shows cleanly that debt is a one-period commitment instrument.
- It also shows that the debt interval is not chosen to maximize shareholder value in general; it is chosen to achieve the minimum discipline needed to prevent removal.

5.3 Proposition 2: the full three-period partition

Read. Proposition 2 is the paper's core result. It partitions managers into five regions, each associated with a different maturity pattern of debt.

Why it matters.

- This is where maturity structure becomes endogenous.
- The proposition delivers a particularly sharp prediction: some relatively better managers issue first-period debt even though worse managers do not, because they need to avoid an *early* takeover but not later discipline.

5.4 Proposition 3: debt frequency in the N-period model

Read. Proposition 3 compresses the N-period problem into the simple integer formula (14).

Why it matters.

- It links debt frequency to the quality of investment opportunities.
- It turns the paper from a 3-period example into a general theory of why some firms should have more frequent debt service than others.

5.5 Proposition 4: stochastic outcomes

Read. Proposition 4 shows that the deterministic logic survives once projects are noisy. The same maturity ranking persists, but debt levels become uniquely pinned down by indifference conditions involving bankruptcy probabilities.

Why it matters.

- The model now predicts equilibrium bankruptcy, not just off-path bankruptcy.
- It delivers precise comparative statics with respect to control benefits, empire-building benefits, and managerial type.

6 Discussion and empirical implications

6.1 Predictions shared with free-cash-flow theories

- Better investment opportunities imply less debt.
- More profitable or faster-growing firms should have less leverage.
- Mature or declining industries should have more leverage.
- If takeover threat rises unexpectedly, leverage may rise as a defense, and firm value may rise with the debt issue.

Interpretation. These are the familiar correlations that make free-cash-flow theories appealing, and Zwiebel's model can generate them too.

6.2 Predictions that sharply differentiate this paper

- **Debt and dividends are complements.** High net debt requires high payout. Managers voluntarily pay dividends because debt only disciplines when cash is not left inside the firm.
- **Tests should look at net debt and payouts jointly.** Gross leverage without payout information can be misleading.
- **The cost of excessive debt is loss of discipline.** When bankruptcy is unavoidable, managers may undertake more bad projects because the future no longer matters.
- **Debt frequency is endogenous.** Firms with fewer good opportunities should have more frequent debt obligations.
- **Managerial career matters.** Debt and bankruptcy risk tend to rise over a manager's tenure, while the final endgame features more empire-building and less restraint.

6.3 Debt, performance, and empire-building

A subtle implication is that the cross-sectional relation among debt, performance, and empire-building is *nonlinear*.

- For managers below replacement level, better type means less debt.
- Above replacement level, managers need no disciplining debt at all.
- Therefore the best firms in equilibrium may display *more* empire-building, not less, because they can afford it and still keep control.

Interpretation. This is one of the most interesting empirical warnings in the paper: equilibrium cross sections need not match within-firm causal effects.

6.4 Legacy building and endgame effects

The model predicts that as retirement approaches:

- debt levels and bankruptcy risk rise for constrained managers;
- bad projects become more likely in the last few periods, because future control rents shrink;
- retiring managers may engage in "legacy building."

Interpretation. This is a genuinely dynamic implication that static capital-structure models do not naturally produce.

7 Short summary

- Zwiebel builds a dynamically consistent agency model in which partially entrenched managers voluntarily choose debt.
- Debt works because bankruptcy lowers entrenchment and increases the chance of replacement.
- Managers choose the minimum net debt needed to commit to reject bad projects in enough future periods to avert takeover.
- The model explains not only leverage levels, but also payout policy, debt frequency, maturity structure, the cost of excessive debt, and how capital structure evolves over managerial careers.

8 Conclusion

8.1 Core conclusion (what the paper establishes)

The paper shows that debt can discipline managers even when managers themselves control financing decisions. The reason is not that debt physically prevents inefficient investment, but that it changes the manager's intertemporal trade-off between empire-building today and control tomorrow.

8.2 Economic intuition

A partially entrenched manager wants to stay in power, but also wants to expand. Debt lets the manager tie his own hands by making bad projects more dangerous. The threat that matters is not illiquidity per se; it is the possibility that bankruptcy strips away entrenchment and invites replacement.

8.3 Takeaway

Zwiebel's main contribution is to turn capital structure into a dynamic governance instrument. Once one views debt as a choice managers make to manage future control threats, dividends, maturity, distress costs, and even late-career empire-building all fit into one coherent story.